DATA SCIENCE LABORATORY MANUAL

Module III ? Data Visualization

Topic: Matplotlib

OVERVIEW & LEARNING OBJECTIVES

Figure architecture, line plots, scatter charts, bar graphs, histograms, customizing axes, legends, and styling.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Matplotlib in modern Data Science workflows.
- Detailed breakdown: Figure architecture, line plots, scatter charts, bar graphs, histograms, customizing axes, legends, and sty
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.